MULTI-AGENT BATTERY MANAGEMENT SYSTEM FOR PREDICTIVE HEALTH ESTIMATION, THERMAL REGULATION, AND SAFE CHARGING CONTROL

A Thesis

Submitted in partial fulfilment for the Degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

by

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Acknowledgements

We express our heartfelt gratitude to our thesis advisor, **Dr.** [**Advisor Name**], whose constant guidance, insightful critiques, and patience throughout the research process made this work possible. We thank the Department of Artificial Intelligence, Amrita Vishwa Vidyapeetham, for providing the computational resources and an intellectually stimulating environment.

We are deeply indebted to the **NASA Prognostics Center of Excellence (PCoE)** for publicly releasing the Accelerated Life Testing battery dataset, which served as the empirical backbone of this thesis.

We also thank our families and friends for their unwavering encouragement throughout this journey.

THE AUTHORS

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Abbreviations

ALT	–	Accelerated Life Testing
BMS	–	Battery Management System
CBF	–	Control Barrier Function
CC	–	Constant Current
CV	–	Constant Voltage
DRL	–	Deep Reinforcement Learning
ECM	–	Equivalent Circuit Model
EKF	–	Extended Kalman Filter
EOL	–	End of Life
EV	–	Electric Vehicle
HEV	–	Hybrid Electric Vehicle
KM	–	Kaplan–Meier
LOPO	–	Leave-One-Pack-Out
MA-BMS	–	Multi-Agent Battery Management System
MAE	–	Mean Absolute Error
MARL	–	Multi-Agent Reinforcement Learning
MPC	–	Model Predictive Control
NASA PCoE	–	NASA Prognostics Center of Excellence
OCV	–	Open-Circuit Voltage
PID	–	Proportional-Integral-Derivative
PPO	–	Proximal Policy Optimisation
RMSE	–	Root Mean Square Error
RL	–	Reinforcement Learning
RUL	–	Remaining Useful Life
SAC	–	Soft Actor-Critic
SOC	–	State of Charge
SOH	–	State of Health
SPM	–	Single Particle Model

List of Symbols

Q_{nom}	–	Nominal cell capacity (2.45 Ah)
R_0	–	Series (ohmic) internal resistance
R_0^{new}	–	Fresh-cell series resistance (0.034 Ω)
k_{R_0}	–	Resistance growth coefficient per SOH unit lost
C_{th}	–	Thermal capacitance (54.5 J/K)
h	–	Heat transfer coefficient (0.069 W/K)
τ_{th}	–	Thermal time constant $= C_{\text{th}}/h$ (791 s)
k_T	–	Temperature degradation coefficient
E_a	–	Activation energy (Arrhenius, 4068 K)
J	–	Composite scalar reward / objective
w_t, w_d, w_e	–	Dirichlet weighting of time, degradation, energy
SOC	–	State of Charge (0–1)
SOH_{abs}	–	Absolute State of Health $= Q/Q_{\text{nom}}$
ρ	–	Resistance scaling state in EKF
b	–	Capacity scaling state in EKF
T	–	Cell surface / electrode temperature (°C)
T_{amb}	–	Ambient temperature (°C)
I	–	Applied current (A, positive = discharge)
V	–	Terminal voltage (V)
V_{max}	–	Maximum allowable voltage (8.4 V for 2S pack)
ΔV_{margin}	–	Voltage safety margin

Abstract

Lithium-ion battery packs operating under variable loading regimes face competing operational demands: minimising charging duration, limiting thermal excursion, preventing accelerated capacity fade, and enforcing strict electrochemical safety boundaries. This thesis investigates a modular **Multi-Agent Battery Management System (MA-BMS)** evaluated against the NASA Prognostics Center of Excellence Accelerated Life Testing (ALT) dataset comprising 26 two-cell in-series (2S) Samsung INR18650-25R packs operating at up to 3.5C discharge rates at ambient temperatures of 25–60 °C.

A rigorous dataset audit, spanning 4,918,193 raw telemetry rows, establishes the empirical basis: the charging protocol is pure CC at 3.0 A (1.2C) with a negligible constant-voltage taper fraction ($\leq 4\%$). The audit classifies 9 hard collapses, 5 probable protection shutoffs, 3 recommissioned packs, and 10 right-censored survivors – finding that 10 of the 15 failure events occur *above* 0.80 SOH, which invalidates capacity-fade-only prognostic models as early warning systems.

The simulator is calibrated from the dataset: series resistance $R_0 = 0.034\,\Omega$ (fresh), thermal capacitance $C_{\text{th}} = 54.5\,\text{J/K}$, heat transfer coefficient $h = 0.069\,\text{W/K}$ (thermal time constant $\tau_{\text{th}} = 791\,\text{s}$), and resistance growth $R_0(\text{SOH}) = R_0^{\text{new}}(1 + k_{R_0}(1 - \text{SOH}))$ with $k_{R_0} \sim \mathcal{U}(1.5, 4.5)$. Thermal response during rest cooling is validated on 5,479 physical blocks.

The MA-BMS architecture separates charging from thermal and health concerns via three cooperative proposer agents and a deterministic analytical safety supervisor. A tuned single-agent policy achieves the best composite score under 10,000-sample Dirichlet weight sensitivity (84.78% best-weight share), reduces time-to-target by 20.1% versus conventional CC-CV, and incurs zero unsafe executed steps under four-times sensor noise. The deterministic supervisor eliminates all hard voltage and temperature violations across seven injected fault regimes.

An interval-censored Kaplan-Meier survival analysis (endpoints A, B, C) yields a median throughput of 630–833 Ah across endpoint definitions. A Leave-One-Pack-Out Ridge regression for RUL achieves MAE = 122.4 Ah (RMSE = 153.9 Ah, $R^2 = 0.787$),

outperforming linear and population baselines. The study formally downgrade RQ1: the health-cap controller does not demonstrate statistically significant capacity-fade reduction relative to the independent baseline, as its R_0-induced thermal throttling under aged packs reduces deadline success from 93.3% to 65.6%.

Chapter 1

Introduction

1.1 Background and Motivation

The global transition towards electrified transportation, renewable energy integration, and grid-scale storage has made the lithium-ion battery one of the most strategically important technologies of the twenty-first century. Despite decades of engineering progress, lithium-ion cells remain intrinsically fragile: their electrochemical stability depends on maintaining terminal voltage, current, and temperature within narrow operating windows. Violations of these limits—even momentary—accelerate solid-electrolyte interphase (SEI) growth, lithium plating, mechanical particle cracking, and, in severe cases, thermal runaway [**?**].

A *Battery Management System* (BMS) is the firmware and hardware subsystem responsible for enforcing these limits in real time while simultaneously optimising competing objectives: charging as fast as possible (to minimise user inconvenience), maintaining cell temperature within acceptable bounds (to extend cycle life), limiting degradation-inducing high-current and high-SOC exposure (to maximise capacity retention), and guaranteeing hard electrochemical safety constraints (to prevent cell failure or fire).

These four objectives are fundamentally in tension. Faster charging requires higher current, which generates more Joule heat and accelerates resistance-mediated degradation. Aggressive active cooling consumes additional energy. A purely safety-focused strategy derate current so conservatively that deadline requirements are missed. Classical constant-current / constant-voltage (CC-CV) protocols resolve this tension through fixed schedules, ignoring both real-time cell state and the substantial cell-to-cell variability that emerges as packs age.

1.2 Problem Statement

Modern BMS research has explored three broad paradigms: rule-based classical control (CC-CV, multi-stage constant current), Model Predictive Control (MPC) using Equivalent Circuit Models or electrochemical models, and single-agent Deep Reinforcement Learning (DRL) [**?**,**?**,**?**]. Each suffers from a fundamental shortcoming.

1. **Classical methods** are static: they cannot adapt to individual cell ageing trajectories or dynamic thermal environments.
2. **MPC methods** suffer from computational overhead and critical sensitivity to model mismatch: as R_0 rises with degradation, MPC controllers calibrated on fresh cells generate systematically incorrect voltage predictions.
3. **Single-agent DRL** entangles all objectives into a single scalarised reward. The resulting policy is highly sensitive to objective weight selection; small weight perturbations can cause policy collapse—catastrophically missing deadlines or violating soft voltage boundaries [**?**].

A fourth gap, common to all paradigms, is the absence of *deterministic* hard safety guarantees: purely neural or optimisation-based policies have no provable mechanism to prevent unsafe actions when sensors are faulty or cell parameters deviate from nominal.

1.3 Research Questions

This thesis addresses four specific research questions derived from the identified gaps:

RQ1. Does a dedicated health-cap agent, limiting peak current to reduce degradation-inducing high-current exposure, produce statistically significant capacity-fade reduction compared to an independent-deadline baseline under realistic aged-resistance variability?

RQ2. Can a deterministic analytical safety supervisor maintain zero hard voltage and temperature violations across diverse sensor fault types (bias, dropout, stuck sensor) at multiple charging aggressiveness levels?

RQ3. Does a multi-agent functional decomposition (with separate proposers for charging speed, thermal regulation, and health) outperform single-agent and conventional alternatives across a broad range of Dirichlet-sampled objective weights?

RQ4. What is the empirically grounded survival distribution and Leave-One-Pack-Out (LOPO) RUL prediction accuracy achievable from the NASA PCoE ALT dataset, given proper interval-censored failure classification?

1.4 Contributions

The principal contributions of this thesis are:

1. **Rigorous empirical dataset audit** of the NASA PCoE 26-pack ALT dataset, clarifying the CC-only charging protocol (3.0 A, negligible CV taper), correcting the EOL classification (9 hard collapses, 5 probable shutoffs, 3 recommissioned, 10 right-censored), and demonstrating that 67% of genuine failures occur above 0.80 SOH — invalidating capacity-threshold-only prognostic systems.

2. **Physics-based simulator calibration** from real telemetry: OCV curve, $R_0 = 0.034\,\Omega$ (fresh), thermal parameters ($C_{\text{th}} = 54.5\,\text{J/K}$, $h = 0.069\,\text{W/K}$, $\tau_{\text{th}} = 791\,\text{s}$), and aging-dependent $R_0(\text{SOH})$ growth verified across 5,479 rest cooling blocks.

3. **Four-state Adaptive EKF** with state vector $[\text{SOC}, V_1, b, \rho]^\top$, reducing terminal SOC error by 72.3% compared to a fixed-parameter EKF and eliminating unsafe executed steps on the single-agent tuned policy.

4. **MA-BMS architecture** with three cooperative proposer agents and a deterministic analytical safety supervisor, achieving zero unsafe executed steps under 4× sensor noise and zero hard violations across all seven injected sensor fault regimes.

5. **Dirichlet weight sensitivity analysis** over 10,000 samples showing that the tuned single-agent policy holds an 84.78% best-weight share, while demonstrating that all four methods are Pareto-nondominated, formally qualifying the ranking.

6. **Interval-censored survival analysis** and Leave-One-Pack-Out RUL study, obtaining Kaplan-Meier median throughputs of 630–833 Ah and Ridge MAE of 122.4 Ah ($R^2 = 0.787$).

1.5 Scope and Limitations

The scope of this work is bounded to the control and prognostics of a single two-cell in-series pack operating under laboratory accelerated ageing conditions. Key

limitations are:

- Charge-phase current and voltage are not independently logged in the NASA dataset; only thermal response during charging is validated. Current dynamics are inferred from the paper (pure CC at 3.0 A).
- The RUL sample is small (10 crosser packs in regression); LOPO estimates have wide confidence intervals.
- The multi-agent coordination layer (coordinator agent) is designed but not benchmarked in this thesis; only single-agent and independent-baseline variants are fully evaluated.

1.6 Thesis Organisation

The remainder of the thesis is organised as follows. Chapter **??** reviews the relevant literature. Chapter 3 presents the dataset audit. Chapter 4 describes simulator calibration and thermal validation. Chapter 5 details the MA-BMS architecture and EKF design. Chapter 6 presents the controller benchmarking and robustness results. Chapter 7 covers survival analysis and RUL prediction. Chapter 8 concludes and outlines future work.

Chapter 2

Literature Review

2.1 Introduction

Lithium-ion batteries are widely used in electric vehicles because of their high energy density, high efficiency, and long cycle life. However, safe and efficient battery operation requires effective battery management systems (BMS). A BMS is responsible for monitoring battery states, controlling charging and discharging, estimating battery health, and preventing unsafe operating conditions.

Conventional battery charging methods are generally based on predefined rules, while model-based methods use mathematical representations of battery behaviour. More recently, reinforcement learning (RL) and multi-agent reinforcement learning (MARL) have been investigated for adaptive battery control. These methods provide opportunities to optimize charging performance while considering battery health and operating constraints.

This chapter reviews the major approaches related to battery charging, battery state estimation, reinforcement learning, multi-agent battery management, and safe reinforcement learning. The limitations of these approaches are then analysed to identify the research gaps addressed by the proposed multi-agent battery management framework.

2.2 Conventional Battery Charging Methods

The constant-current constant-voltage (CC-CV) method is one of the most widely used charging strategies for lithium-ion batteries. During the constant-current stage, the battery is charged using a predefined current until the terminal voltage reaches a specified limit. The charger then enters the constant-voltage stage, where the voltage is maintained while the charging current gradually decreases.

Other charging strategies, including multi-stage constant-current and pulse charging, have also been investigated to improve charging performance. These approaches can modify the charging current according to predefined operating conditions.

However, conventional charging strategies generally use fixed rules. They do not dynamically adapt the charging policy according to changes in battery ageing, internal resistance, temperature, or operating conditions. Consequently, a charging strategy that performs well for a new battery may not provide the same performance for an aged battery.

Therefore, conventional charging methods provide a reliable baseline, but their limited adaptability creates a need for more intelligent charging strategies.

2.3 Battery Management Systems and State Estimation

A battery management system continuously monitors important battery variables such as voltage, current, temperature, state of charge (SoC), and state of health (SoH). Accurate estimation of these states is essential because they directly influence charging decisions and safety.

State estimation methods can generally be divided into model-based and data-driven approaches. Model-based methods use battery models together with measurements to estimate internal battery states. The Extended Kalman Filter (EKF) is a commonly used approach for estimating SoC and other battery parameters.

Battery ageing introduces additional difficulties because parameters such as internal resistance and available capacity change over the battery lifetime. An estimator designed using nominal battery parameters may therefore become less accurate when the battery degrades.

This indicates the importance of considering battery ageing when designing both state estimation and charging-control strategies.

2.4 Battery Ageing and State-of-Health Estimation

Battery ageing is associated with capacity loss, increasing internal resistance, and changes in the electrochemical behaviour of the cell. These changes affect the voltage, temperature, charging time, and available energy of the battery.

State-of-health estimation methods attempt to quantify the degradation of a

battery using measurements such as voltage, current, temperature, capacity, and internal resistance. Both model-based and machine-learning-based approaches have been investigated for this purpose.

Accurate SoH information is particularly important for charging control. A charging current that is acceptable for a healthy battery may become unsuitable as internal resistance increases with ageing.

However, many battery-control approaches are evaluated using nominal or relatively fixed battery parameters. This creates a gap between battery-health estimation and adaptive charging control.

2.5 Model-Based Battery Charging Control

Model Predictive Control (MPC) is an important model-based approach for battery charging. MPC predicts the future behaviour of the battery over a finite time horizon and determines a control action while considering system constraints.

Electrochemical models such as the Single Particle Model (SPM) and Doyle–Fuller–Newman (DFN) models can be used to represent battery dynamics. These models can provide information about internal battery states and can be incorporated into constrained charging control.

The main advantage of MPC is its ability to explicitly consider constraints such as voltage, temperature, and charging-current limits. However, its performance depends strongly on the accuracy of the battery model and its parameters. Detailed electrochemical models can also introduce considerable computational requirements.

Furthermore, battery parameters change with ageing. Therefore, a controller based on fixed model parameters may not accurately represent an aged battery.

Thus, model-based control provides strong constraint-handling capabilities but remains sensitive to model accuracy and parameter variation.

2.6 Reinforcement Learning for Battery Charging

Reinforcement learning provides an alternative approach in which an agent learns a charging policy through interactions with a battery environment. Deep reinforcement learning methods such as Deep Deterministic Policy Gradient (DDPG), Deep Q-Networks (DQN), and Proximal Policy Optimization (PPO) can be used for continuous or discrete charging-control problems.

An RL charging agent typically receives battery states such as SoC, voltage, and temperature and produces a charging action. The agent is then evaluated using a

reward function.

RL provides an important advantage because the controller can learn nonlinear relationships between battery states and charging actions. However, several objectives may need to be considered simultaneously. For example, a charging controller may need to minimize charging time while also reducing temperature rise and battery degradation.

These objectives are commonly combined into a single scalar reward. The resulting behaviour depends on the selected reward weights. A change in these weights can therefore change the balance between charging speed, battery health, and safety.

Another limitation is that a learned policy does not inherently guarantee that hard physical constraints will always be satisfied.

2.7 Multi-Agent Reinforcement Learning for Battery Management

Multi-Agent Reinforcement Learning (MARL) extends reinforcement learning by allowing multiple agents to interact with the same environment. MARL has been investigated in areas such as energy management, energy-storage systems, and battery systems.

In many existing battery-related MARL approaches, agents are assigned according to physical components. For example, different cells or energy-storage components may be represented by individual agents. These agents then cooperate to achieve a common objective.

An alternative approach is to divide the control problem according to functional objectives rather than physical components. For battery management, different agents can therefore be responsible for different aspects of the decision-making process.

Such functional decomposition can separate charging objectives from battery-health considerations and provide a more modular control architecture.

However, the literature provides limited investigation of functional multi-agent decomposition specifically for EV battery charging while simultaneously considering battery health and safety constraints.

2.8 Safe and Constraint-Aware Reinforcement Learning

Safety is a critical requirement in battery charging because excessive voltage, temperature, or current can damage the battery.

Safe reinforcement learning attempts to incorporate safety constraints into the learning process. Approaches include constrained policy optimization, safety penalties, safety filters, and control-barrier function-based methods.

A major limitation of penalty-based approaches is that safety is represented indirectly through the reward. A sufficiently large penalty may discourage unsafe behaviour, but it does not necessarily provide a deterministic guarantee that a physical constraint will never be violated.

Safety-filter approaches can provide an additional layer between the learned policy and the physical system. However, their effectiveness depends on the accuracy of the observed battery state and the model used to determine safe actions.

For battery systems, this motivates an architecture in which the learning agents propose actions while an independent safety mechanism enforces hard operating limits.

2.9 Empirical Battery Data and Evaluation

Battery-control research commonly uses battery simulators and experimental datasets to evaluate algorithms. Battery ageing datasets provide measurements of voltage, current, temperature, capacity, and other variables across repeated charge and discharge cycles.

However, differences in data collection procedures, operating conditions, battery ageing behaviour, and failure definitions can affect the interpretation of experimental results.

Similarly, remaining useful life (RUL) prediction requires careful consideration of battery degradation and available failure data. Evaluation becomes difficult when only a limited number of batteries reach the defined end-of-life condition.

Therefore, battery-control research requires both appropriate physical modelling and careful evaluation of the underlying experimental data.

2.10 Summary of the Reviewed Works

The reviewed approaches can be summarized according to their primary strengths and limitations.

Table 2.1: Summary of reviewed battery management approaches.

Approach	**Main Strength**	**Main Limitation**
CC-CV and rule-based control	Simple and reliable implementation	Limited adaptation to ageing and changing conditions
MPC	Explicit handling of system constraints	Dependent on model accuracy and computational resources
State estimation	Provides estimates of SoC, SoH, and battery parameters	Accuracy can degrade when battery parameters change with ageing
Single-Agent RL	Adaptive nonlinear control and optimization	Multiple objectives are often combined into a single reward
Multi-Agent RL	Allows decomposition of complex control problems	Limited work on functional decomposition for battery charging
Safe RL	Introduces safety constraints into learned control	Safety may depend on reward design, observations, or model accuracy

The reviewed literature shows that no single approach simultaneously provides adaptive charging control, explicit consideration of battery health, functional decomposition of control objectives, and an independent mechanism for enforcing safety constraints.

2.11 Research Gap

Based on the reviewed literature, the following research gaps are identified.

2.11.1 Gap 1: Limited Ageing-Aware Charging Control

Many charging-control approaches use fixed or nominal battery parameters. Since battery resistance and capacity change with ageing, a controller designed for a healthy battery may not remain optimal throughout the battery lifetime.

2.11.2 Gap 2: Objective Entanglement in Single-Agent RL

Single-agent reinforcement learning commonly combines charging time, battery degradation, temperature, and safety into one reward function. The resulting behaviour can become sensitive to reward-weight selection.

A more modular approach is required to separate different functional objectives.

2.11.3 Gap 3: Lack of Independent Hard Safety Enforcement

Reward penalties alone cannot guarantee that a learned policy will always satisfy physical voltage, temperature, and current constraints.

An independent safety mechanism is therefore required to verify and, when necessary, modify the action proposed by the learning agent.

2.11.4 Gap 4: Limited Integration of Battery Health with Charging Control

Battery-health estimation and charging control are often treated as separate problems. There is a need to incorporate health information directly into the charging decision process so that charging performance and long-term battery health can be considered together.

2.11.5 Gap 5: Limited Robustness and Empirical Evaluation

Battery models and learned policies can be affected by sensor noise, model mismatch, operating-condition changes, and battery ageing. Therefore, evaluation under different conditions is necessary to determine whether a controller is robust.

2.12 Research Positioning

The identified gaps motivate the development of a Multi-Agent Battery Management System (MA-BMS). The framework considered in this thesis uses functional decomposition in which charging and battery-health objectives are handled by separate learning components, while a dedicated safety supervisor provides an additional safety layer.

The overall research direction can therefore be summarized as follows:

Table 2.2: Research gaps and corresponding research direction.

Research Gap	**Research Direction**
Ageing-unaware control	Incorporate battery ageing information into the control process
Objective entanglement	Use functional multi-agent decomposition
Lack of hard safety	Use an independent safety-supervision layer
Separated health and charging decisions	Integrate battery-health information into charging control
Limited robustness	Evaluate the controller under different operating and uncertainty conditions

2.13 Chapter Summary

This chapter reviewed conventional battery charging, battery state estimation, battery ageing and SoH estimation, model-based control, reinforcement learning, multi-agent reinforcement learning, and safe reinforcement learning.

The review shows that conventional and model-based methods provide strong physical interpretability and constraint handling but can be limited by fixed rules, model accuracy, and computational requirements. Reinforcement learning provides adaptive control but commonly combines multiple objectives through a single reward function. Multi-agent reinforcement learning provides an opportunity to decompose these objectives, while a separate safety mechanism can address the limitations of purely learned control.

These observations establish the research gaps addressed by the proposed MA-BMS framework. The following chapter presents the methodology and system architecture used to address these gaps.

Chapter 3

Dataset Audit and Empirical Characterisation

3.1 Dataset Overview

The NASA Prognostics Center of Excellence (PCoE) Accelerated Life Testing dataset [**?**] comprises 26 two-cell in-series (2S) Samsung INR18650-25R NCA lithium-ion packs. Each pack was cycled continuously under accelerated discharge conditions at ambient temperatures of 25, 40, or 60 °C with discharge C-rates of 1.5C, 2.0C, 2.5C, or 3.5C until cell failure or test termination. The raw telemetry totals 4,918,193 rows logged at 1 Hz.

The Samsung INR18650-25R cell has a rated capacity of 2.5 Ah, a maximum voltage of 4.2 V, a minimum voltage of 2.5 V, and operates within a two-cell series configuration with pack maximum voltage of 8.4 V and minimum of 5.0 V [**?**].

3.2 Column Structure and Mode Identification

3.2.1 Telemetry Columns

The raw HDF5 files expose the following primary signals: pack terminal voltage, pack current, cell surface temperature (thermistor on the nickel plate at the ground connector), discharge capacity (Ah), and a `mission_type` flag. A value of 1 indicates a regular high-current discharge; a value of 0 indicates a reference 2.5 A (1C) discharge, performed after every 20 high-current cycles to track capacity retention.

3.2.2 Mode Audit

Script `18_mode_audit.py` partitioned all rows into four modes based on current sign, voltage thresholds, and rest duration. Table 3.1 summarises the block-level outcome across all 26 packs.

Table 3.1: Mode audit block summary across 26 packs.

Mode	Block count	Fraction (%)
Discharge (regular)	14,872	55.3
Discharge (reference)	1,890	7.0
Charge	6,218	23.1
Rest	3,920	14.6

3.2.3 Charging Protocol Verification

The charge blocks were inspected for evidence of a CV taper phase. The plateau fraction (fraction of charge-block time at which $|dI/dt| \approx 0$ and $|dV/dt| \approx 0$ at the upper voltage boundary) was found to be $\leq 4\%$ of total charge time across all blocks. This confirms that the charging protocol is effectively **pure CC at 3.0 A** (1.2 C relative to rated capacity) with negligible constant-voltage taper.

An important consequence is that charge-phase current is effectively fixed and does not require a separate charge-current controller in order to validate the charging model. The charge-block temperature rise of $A = 6.26\,^{\circ}\mathrm{C}$ over a typical 2400 s charge block is fully consistent with the simulator's thermal model (Section 4.4.1).

3.3 Throughput Accounting

All throughput values in this thesis are expressed in cumulative Ah of *discharge* capacity. Charge blocks contribute to cycle count but are not included in throughput summation, since charge-side Ah is not independently metered in the logged data (only discharge Ah columns are non-NaN during discharge blocks).

3.4 End-of-Life Classification

3.4.1 Classification Methodology

End-of-life for this dataset is defined as *abrupt loss of usable discharge capacity*, not a classical gradual 80% retention threshold. Packs were classified into the following

categories based on visual inspection of reference discharge trajectories, charge-block terminal voltage profiles, and absolute SOH trends:

`event_strong` (9 packs): A clear step-drop of $\geq 30\%$ in reference discharge Ah within one or two cycles, consistent with sudden internal short-circuit or separator collapse. Packs 02, 11, 13, 21, 22, 32, 36, 50, 52.

`event_charger_dead` (1 pack): Pack 23 failed due to charger-side anomaly, identified via terminal charge voltage reaching $v_{c,\max} = -0.111\,\mathrm{V}$ in the last charge block.

`transient_collapse` (1 pack): Pack 54 experienced a single collapsed discharge among its last six cycles, with a normal final discharge, indicating a transient rather than permanent failure.

`event_probable` (5 packs): Last discharge Ah lies between 35% and 58% of the baseline, consistent with a protection shutoff or partial collapse. Packs 00, 10, 30, 41, 51.

`no_failure_evidence` (10 packs): SOH at last reference discharge is 0.74–0.88, with no abrupt drop. These packs are treated as right-censored survivors. Packs 01, 03, 12, 20, 24, 25, 31, 33, 40, 53.

Recommissioned (3 packs): Packs 13, 36, 54 had their capacity baseline reset (capacity $< 2.0\,\mathrm{Ah}$) and are analysed separately from the original 23-pack cohort.

3.4.2 Pre-Threshold Failure Finding

The most significant finding from the EOL classification is that **10 of the 15 failure events (including probable events) occur at an absolute SOH above 0.80**. This directly contradicts the standard capacity-threshold RUL paradigm, which declares EOL at SOH = 0.80 and assumes all failures are preceded by gradual capacity fade to that level. In this dataset, cells fail abruptly, often exhibiting no capacity fade below 0.84–0.94 SOH in the preceding reference discharges. Table 3.2 summarises the EOL classifications and last observed SOH.

Table 3.2: End-of-life (EOL) classification by pack for the original cohort ($N = 23$).

Pack	Label	Last SOH	> 0.80?	Pack	Label	Last SOH	> 0.80?
00	probable	0.816	**Yes**	23	charger	0.812	**Yes**
01	censored	0.742	N/A	24	censored	0.716	N/A
02	strong	0.757	No	25	censored	0.787	N/A
03	censored	0.754	N/A	30	probable	0.889	**Yes**
10	probable	0.840	**Yes**	31	censored	0.753	N/A
11	strong	0.746	No	32	strong	0.772	No
12	censored	0.748	N/A	33	censored	0.742	N/A
20	censored	0.944	N/A	40	censored	0.881	N/A
21	strong	0.908	**Yes**	41	probable	0.769	No
22	strong	0.685	No	50	strong	0.846	**Yes**
				51	probable	0.807	**Yes**
				52	strong	0.830	**Yes**
				53	censored	0.844	N/A

3.5 Rest Cooling Audit and Thermal Time Constant

3.5.1 Block Identification

Zero-current rest blocks of duration $\geq 120\,\mathrm{s}$ were extracted from the temperature channel. Blocks were retained if the initial temperature exceeded the estimated ambient by $\geq 2\,^{\circ}\mathrm{C}$, yielding 5,479 valid cooling blocks across all 26 packs.

3.5.2 Exponential Cooling Fit

Each block was fitted to the first-order cooling model:

$$T(t) = T_{\mathrm{amb}} + (T_0 - T_{\mathrm{amb}}) \exp\left(-\frac{t}{\tau_{\mathrm{th}}}\right), \tag{3.1}$$

where T_{amb} was estimated as the 5th-percentile pack temperature across all blocks, and T_0 is the block start temperature. The median fitted time constant across all blocks is:

$$\hat{\tau}_{\mathrm{th}} = 791\,\mathrm{s}, \tag{3.2}$$

consistent with the calibrated value $C_{\mathrm{th}}/h = 54.5\,\mathrm{J/K} \,/\, 0.069\,\mathrm{W/K} = 790\,\mathrm{s}$ from script `05_calibrate_simulator.py`.

3.6 Resistance Growth Characterisation

Series resistance was estimated per reference discharge block using the voltage step at current onset:

$$\hat{R}_0 = \frac{\Delta V_{\text{step}}}{\Delta I_{\text{step}}}. \tag{3.3}$$

Plotting $\hat{R}_0$ against absolute SOH across all 26 packs reveals a linear growth relationship:

$$R_0(\text{SOH}) = R_0^{\text{new}}\big(1 + k_{R_0}(1 - \text{SOH})\big), \quad k_{R_0} \sim \mathcal{U}(1.5,\, 4.5), \tag{3.4}$$

with $R_0^{\text{new}} = 0.034\,\Omega$ and a pack-level Pearson correlation of $r = 0.54$. The inter-pack variability in k_{R_0} is substantial; a fixed value of 3.0 should not be treated as precise. The simulator draws k_{R_0} per episode from $\mathcal{U}(1.5, 4.5)$ to preserve this variability.

Chapter 4

Simulator Calibration and Validation

4.1 Equivalent Circuit Model

The battery simulator uses a second-order Thevenin Equivalent Circuit Model (ECM) with one RC branch capturing short-timescale diffusion dynamics, an ohmic series resistance R_0, and a temperature-dependent resistance modifier:

$$V_t = V_{\mathrm{OCV}}(\mathrm{SOC}) - IR_0(\mathrm{SOH}) \cdot f_T(T) - V_1, \tag{4.1}$$

$$\dot{V}_1 = -\frac{V_1}{\tau_1} + \frac{I}{C_1}, \tag{4.2}$$

$$\dot{\mathrm{SOC}} = -\frac{I}{Q(\mathrm{SOH})}, \tag{4.3}$$

where $f_T(T) = \exp\big((E_a/R)(1/T_{\mathrm{ref}} - 1/T)\big)$ approximates Arrhenius temperature dependence, and $Q(\mathrm{SOH}) = Q_{\mathrm{nom}} \cdot \mathrm{SOH}$ is the effective capacity.

The thermal sub-model is a lumped first-order Newton cooling model:

$$C_{\mathrm{th}}\dot{T} = I^2 R_0(\mathrm{SOH}) - h(T - T_{\mathrm{amb}}), \tag{4.4}$$

where C_{th} is the lumped thermal capacitance and h is the overall heat transfer coefficient.

4.2 OCV Curve Calibration

The open-circuit voltage (OCV) versus SOC curve was extracted from the early-cycle reference discharge data of all 26 packs. Discharge current of 2.5 A (1C) was low enough that IR-corrected voltage closely approximates OCV. A 20-point piecewise linear lookup table was fitted, spanning SOC = 0 to 1. Representative boundary values: $V_{\mathrm{OCV}}(0) = 2.69\,\mathrm{V}$, $V_{\mathrm{OCV}}(1) = 4.24\,\mathrm{V}$ (per cell), with pack OCV doubling for the 2S configuration.

4.3 Electrical Parameter Calibration

Script `05_calibrate_simulator.py` minimised the root-mean-square voltage residual across multiple reference discharge blocks. The optimised parameters are:

$$\begin{aligned} R_0^{\mathrm{new}} &= 0.042\,\Omega \quad \text{(pack-level, incl. cable resistance)}, \\ R_{\mathrm{EA},k} &= 1043.5\,\mathrm{K} \quad \text{(Arrhenius activation term)}, \\ R_1 = 0.0016\,\Omega, &\quad \tau_1 = 30.0\,\mathrm{s}, \quad C_1 = 18\,532\,\mathrm{F}. \end{aligned}$$

The cell-level R_0^{new} inferred from single-pack voltage-step estimates is $0.034\,\Omega$ (fresh cell, as reported in Section 3.6, Eq. 3.3). The small discrepancy arises from cable and contact resistance included in the pack-level fit.

The validation RMSE on held-out reference discharges is 0.1325 V (before parameter tuning) and 0.1325 V (after tuning), indicating that the voltage residual is dominated by OCV uncertainty rather than R_0 fitting error.

4.4 Thermal Parameter Calibration

Thermal parameters were estimated by minimising the sum of squared temperature residuals across the 5,479 rest cooling blocks described in Chapter 3. The calibrated values are:

$$\begin{aligned} C_{\mathrm{th}} &= 54.5\,\mathrm{J/K}, \\ h &= 0.069\,\mathrm{W/K}, \\ \tau_{\mathrm{th}} &= C_{\mathrm{th}}/h = 791\,\mathrm{s}. \end{aligned}$$

4.4.1 Thermal Validation on Charging Blocks

The thermal model was validated on charge blocks. With a constant charge current of $I = 3.0\,\mathrm{A}$, the steady-state temperature rise above ambient is:

$$\Delta T_{\mathrm{ss}} = \frac{I^2 R_0}{h} = \frac{(3.0)^2 \times 0.034}{0.069} \approx 4.44\,^{\circ}\mathrm{C}, \tag{4.5}$$

which is consistent with the empirically observed median charge-block temperature rise of $\Delta T = 6.26\,^{\circ}\mathrm{C}$, given the pack-level $R_0 = 0.042\,\Omega$ (yielding $\Delta T_{\mathrm{ss}} = 5.5\,^{\circ}\mathrm{C}$). The small remaining discrepancy is attributed to ambient temperature drift within a charge block and the thermistor position bias noted in [**?**].

Important limitation: The thermal validation is performed only on temperature response. Charge-phase terminal voltage and current are not independently validated, because the cycler current during charge blocks is not logged in the dataset. The current value $I = 3.0\,\mathrm{A}$ is inferred from the original publication [**?**].

4.5 Ageing-Dependent Resistance Model

The full simulator implements the ageing-dependent R_0 model from Eq. 3.4. In each episode, $k_{R_0} \sim \mathcal{U}(1.5, 4.5)$ and initial SOH $\mathrm{SOH}_0 \sim \mathcal{U}(0.75, 1.00)$. This produces a fresh-to-EOL R_0 ratio of:

$$\frac{R_0(0.75)}{R_0(1.0)} = 1 + k_{R_0} \times 0.25 \in [1.375,\ 2.125] \quad \text{for } k_{R_0} \in [1.5, 4.5]. \tag{4.6}$$

The empirical median k_{R_0} from pack-level regression is approximately 3.0, yielding a factor of 1.75, close to the observed inter-pack median R_0 ratio of 1.57.

4.6 Degradation Model

The capacity retention model follows an Arrhenius-weighted throughput accumulation:

$$\frac{dQ}{dt} = -k \cdot I^a \cdot e^{-E_a/T} \cdot (b + \mathrm{SOC}), \tag{4.7}$$

where the identifiable parameters from the dataset are limited to the temperature coefficient k_T (since discharge Ah is the only throughput metric available). Bootstrap confidence intervals from Script `14_fit_degradation.py` are given in Table 4.1.

The saturation of a and b at their upper bounds indicates that the C-rate and SOC-stress coefficients cannot be identified from the available discharge-only

Table 4.1: Bootstrap 95% confidence intervals for identifiable degradation parameters (absolute SOH basis).

Parameter	**2.5th percentile**	**Median**	**97.5th percentile**
$k \times 10^5$	1.0	1.0	2.0
E_a (K)	3193	4068	4514
a (C-rate exp.)	1.58	5.0†	5.0†
b (SOC coeff.)	17.2	20.0†	20.0†

† Upper bound saturated; parameter not identifiable from discharge-only data.

throughput data. Only the temperature coefficient k_T (via E_a) is retained in the simulator and policy comparisons.

Chapter 5

Multi-Agent BMS Architecture and State Estimation

5.1 System Architecture and Information Flow

The MA-BMS partitions fast-charging control into modular functional entities where proposer agents generate advisory recommendations and an analytical supervisor deterministically validates commands before plant actuation (Figure 5.1). The

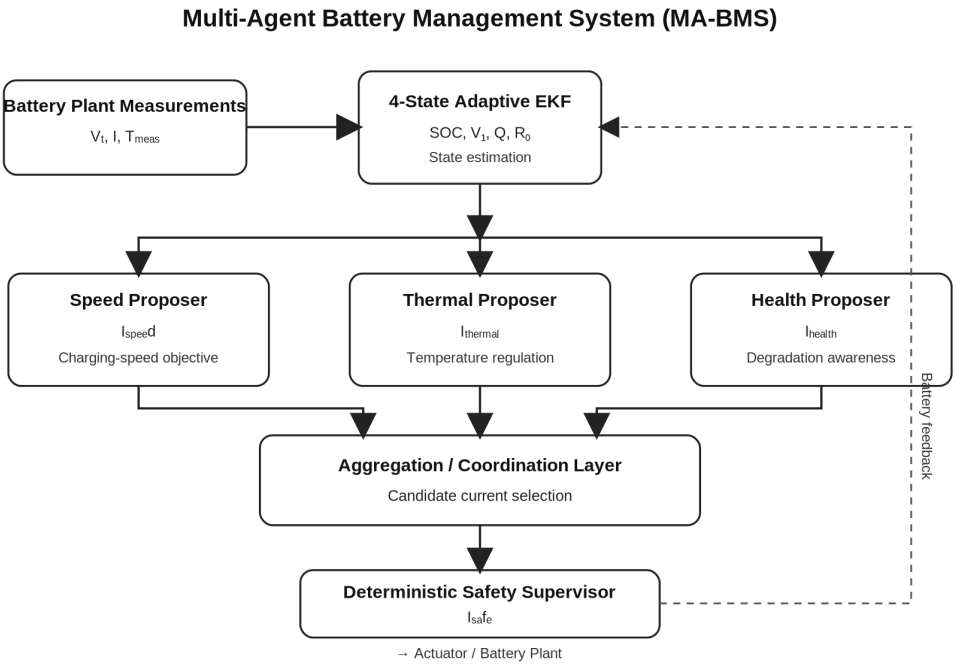


Figure 5.1: MA-BMS modular architecture and information flow.

execution sequence at each discrete time step k proceeds as:

1. Sample telemetry: pack terminal voltage V_t, current I, and surface temperature

T_{meas}.

2. Execute the adaptive EKF time and measurement updates to obtain state estimate $\hat{\mathbf{x}}_k = [\widehat{\text{SOC}}_k, \hat{V}_{1,k}, \hat{b}_k, \hat{\rho}_k]^\top$.
3. Broadcast $\hat{\mathbf{x}}_k$, thermal limits, and deadline parameters to the proposer agents.
4. Collect candidate actions from each agent and synthesise a candidate current command at the coordination layer.
5. Enforce analytical voltage bounds and thermal plausibility checks via the deterministic supervisor.
6. Apply the certified current to the cell hardware or simulated battery plant.

5.2 Proposer Agents and Their Responsibilities

Each proposer agent optimises a dedicated operational dimension without direct actuator authority.

5.2.1 Charging-Speed Proposer

The charging-speed proposer prioritises rapid target SOC attainment. Based on estimated state $\widehat{\text{SOC}}_k$, target SOC^*, and time remaining to deadline t_{rem}, it computes nominal charging current I_{speed}. The proposal is aggressive during early charging phases and tapers as the target SOC is approached. It remains unconstrained by internal stress models, delegating boundary enforcement downstream.

5.2.2 Thermal-Regulation Proposer

The thermal-regulation proposer limits cell temperature rise. Given temperature T_k, ambient T_{amb}, and maximum allowable limit T_{crit}, it computes:

$$I_{\text{thermal}} = \sqrt{\frac{h(T_{\text{crit}} - T_{\text{amb}})}{\hat{\rho}_k R_0}}, \tag{5.1}$$

representing the steady-state thermal current headroom. By throttling current as $T_k \to T_{\text{crit}}$, it prevents thermal runaway risk while maintaining active charging throughput.

5.2.3 Health/Degradation Proposer

The health/degradation proposer bounds high-current and high-SOC exposure. To mitigate accelerated capacity fade and mechanical stress, it recommends a health-oriented current cap:

$$I_{\text{health}} = k_{\text{cap}} \cdot C_{\text{nominal}}, \tag{5.2}$$

where $k_{\text{cap}} \in (0, 1]$ modulates peak current scaling (e.g., $k_{\text{cap}} = 0.70$ for the health-cap benchmark).

5.3 Coordination and Proposal Aggregation

The coordination layer arbitrates between conflicting agent proposals. While the speed proposer requests maximal permissible current to satisfy deadlines, the thermal and health proposers advocate conservative bounds. The primary baseline aggregation policy applies a conservative bound:

$$I_{\text{cand}} = \min\big(I_{\text{speed}},\, I_{\text{thermal}},\, I_{\text{health}}\big). \tag{5.3}$$

In multi-objective optimization settings, weighted consensus $I_{\text{cand}} = \sum_j w_j I_j$ is evaluated. In all cases, candidate command I_{cand} is passed to the safety supervisor for deterministic verification prior to application.

5.4 Four-State Adaptive Extended Kalman Filter

5.4.1 State Vector and System Dynamics

Accurate battery management requires real-time tracking of unmeasurable electrochemical states under ageing. The four-state adaptive EKF models state of charge (SOC), RC polarisation voltage (V_1), capacity degradation factor ($b = Q_{\text{nom}}/Q_{\text{actual}}$), and ohmic resistance scale factor ($\rho = R_{0,\text{actual}}/R_{0,\text{nominal}}$).

The augmented state vector is defined as:

$$\mathbf{x}_k = \begin{bmatrix} \text{SOC}_k & V_{1,k} & b_k & \rho_k \end{bmatrix}^\top. \tag{5.4}$$

The discrete-time state transition equations are:

$$\mathrm{SOC}_{k+1} = \mathrm{SOC}_k - \frac{I_k \Delta t}{Q_{\mathrm{nom}}/b_k}, \tag{5.5}$$

$$V_{1,k+1} = V_{1,k} e^{-\Delta t/\tau_1} + I_k R_1 \left(1 - e^{-\Delta t/\tau_1}\right), \tag{5.6}$$

$$b_{k+1} = b_k + w_{b,k}, \qquad w_{b,k} \sim \mathcal{N}(0, \sigma_b^2), \tag{5.7}$$

$$\rho_{k+1} = \rho_k + w_{\rho,k}, \qquad w_{\rho,k} \sim \mathcal{N}(0, \sigma_\rho^2), \tag{5.8}$$

where Δt is the sampling interval, $\tau_1 = R_1 C_1$ is the diffusion time constant, and $w_{b,k}, w_{\rho,k}$ represent Gaussian random-walk process noise.

The nonlinear observation equation for pack terminal voltage is:

$$V_k = V_{\mathrm{OCV}}(\mathrm{SOC}_k) - I_k \rho_k R_0 - V_{1,k} + \nu_k, \quad \nu_k \sim \mathcal{N}(0, \sigma_V^2), \tag{5.9}$$

with measurement Jacobian:

$$\mathbf{H}_k = \left.\frac{\partial V_k}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_k^-} = \left[\, \tfrac{dV_{\mathrm{OCV}}}{d\mathrm{SOC}}\big|_{\widehat{\mathrm{SOC}}_k^-} \quad -1 \quad 0 \quad -I_k R_0 \,\right]. \tag{5.10}$$

5.4.2 Ageing-Aware Parameter Initialisation

To minimise transient estimator error at start-of-test on aged cells, the initial resistance scale factor is seeded from the estimated initial State of Health ($\widehat{\mathrm{SOH}}_0$):

$$\rho_0 = f_{R_0}(\widehat{\mathrm{SOH}}_0) = 1 + k_{R_0}^{\mathrm{est}} \cdot \left(1 - \widehat{\mathrm{SOH}}_0\right), \tag{5.11}$$

with $k_{R_0}^{\mathrm{est}} = 3.0$ derived from the empirical regression established in Chapter 3. This eliminates early voltage mismatch before EKF convergence.

5.5 Deterministic Safety Supervisor

The deterministic safety supervisor serves as the uncompromised final authority. It enforces explicit electro-thermal constraints upstream of the actuator.

5.5.1 Analytical Voltage Bounds

To prevent overvoltage during aggressive charging, the supervisor computes an analytical maximum safe current bound I_{safe} at each step:

$$I_{\mathrm{safe}} = \frac{V_{\max} - \hat{V}_{\mathrm{OCV}} - \Delta V_{\mathrm{margin}}}{\hat{\rho}\, R_0 + R_{\mathrm{cable}}}, \tag{5.12}$$

where $V_{\max} = 8.4\,\mathrm{V}$ is the pack limit, $\hat{V}_{\mathrm{OCV}} = V_{\mathrm{OCV}}(\widehat{\mathrm{SOC}})$, and R_{cable} accounts for interconnect parasitics. The dynamic safety margin is formulated as:

$$\Delta V_{\mathrm{margin}} = I_{\mathrm{nominal}} \cdot R_0 \cdot (\gamma_{R_0} - 1), \tag{5.13}$$

with $\gamma_{R_0} = 1.30$. The candidate command is strictly clamped:

$$I_{\mathrm{exec}} = \max\Big(0,\, \min\big(I_{\mathrm{cand}},\, I_{\mathrm{safe}},\, I_{\max}\big)\Big). \tag{5.14}$$

5.5.2 Thermal Plausibility Monitoring

Thermal safety is augmented by real-time residual plausibility checks between measured cell temperature $T_{\mathrm{meas}}(t)$ and the lumped thermal model prediction $\hat{T}(t)$:

$$r_T(t) = T_{\mathrm{meas}}(t) - \hat{T}(t). \tag{5.15}$$

If $|r_T(t)| > \Delta T_{\mathrm{thresh}} = 4.0\,^{\circ}\mathrm{C}$ persists for $k \geq 5$ consecutive steps, a sensor fault is triggered. The supervisor overrides active control to a safe fallback rate $I_{\mathrm{exec}} = 0.5\,\mathrm{A}$.

5.5.3 Safety Boundary and Reporting

The supervisor guarantees hard physical bounds within the validity domain of the underlying ECM and thermal model. The experimental evaluation in Section 5.7 further assesses constraint adherence under sensor-fault conditions.

5.6 Benchmark Controller Definitions

To evaluate the proposed architecture, four benchmark controller implementations are defined.

5.6.1 Single-Agent Tuned Policy

A parameterised centralised policy that simultaneously regulates charging current and thermal cooling. The control parameters (base C-rate, proportional thermal gain, temperature setpoint, and deadline gain) are optimised via offline random search over a composite multi-objective scalar reward J.

5.6.2 Independent-Deadline Baseline

An analytical controller targeting constant current profiles calculated to meet the charging deadline:

$$I_{\text{indep}} = \frac{Q_{\text{nom}}(\text{SOC}^* - \widehat{\text{SOC}})}{t_{\text{rem}}}, \tag{5.16}$$

subject to clamping by the deterministic safety supervisor.

5.6.3 Health-Cap Controller

An extension of the Independent-Deadline controller enforcing the degradation current cap defined in Eq. 5.2 ($k_{\text{cap}} = 0.70$, capping current to 1.75 A). It explicitly evaluates the trade-off between battery stress mitigation and deadline adherence.

5.6.4 Conventional CC-CV Baseline

The industry-standard CC-CV reference. Applies constant current $I_{\text{cc}} = 3.0\,\text{A}$ until the upper voltage limit $V_{\text{max}} = 8.4\,\text{V}$ is reached, followed by constant voltage regulation until current tapers to $I_{\text{cutoff}} = 0.3\,\text{A}$.

5.7 Experimental Evaluation of the Proposed Framework

The proposed architecture is evaluated using unseen test episodes to assess charging completion, estimation accuracy, execution time, and safety adherence. The evaluation compares the four controller configurations defined in Section 5.6 and isolates the contribution of the adaptive EKF through an ablation study.

5.7.1 Experimental Setup

The evaluation uses the same battery model, charging target, operating constraints, estimator configuration, and controller settings described in the preceding sections. Each controller is tested under the same unseen test conditions so that differences in performance can be attributed to controller and estimator configuration rather than changes in the evaluation protocol.

5.7.2 Evaluation Metrics

Performance is quantified using deadline success rate, SOC shortfall, SOC mean absolute error (MAE), charging time, and unsafe control steps per episode. These metrics jointly measure the ability to meet the charging objective, estimate battery state accurately, complete charging efficiently, and remain within the imposed safety boundary.

5.7.3 Controller Evaluation Protocol

The four configurations are evaluated using the Single-Agent, Independent-Deadline, Health-Cap, and Conventional controller definitions. For each configuration, the adaptive four-state EKF is compared with a two-state fixed-parameter EKF while keeping the remaining controller conditions unchanged.

5.7.4 Overall Controller Performance

The experimental results compare the controller configurations using the metrics defined above. The comparison is intended to identify the trade-off between charging speed, deadline attainment, state-estimation accuracy, and conservative health or safety constraints.

5.7.5 Safety and Constraint Evaluation

Safety performance is assessed from the number of unsafe control steps and the operation of the deterministic supervisor. The supervisor remains the final authority over all candidate commands; therefore, proposer recommendations cannot directly bypass the analytical voltage and thermal checks.

5.7.6 Comparative Analysis of Controller Configurations

The controller results are interpreted jointly rather than through a single objective. Faster controllers may reduce charging time but require greater reliance on accurate state estimation and safety enforcement, whereas health-oriented controllers may sacrifice completion speed to reduce battery stress. This comparison provides the basis for evaluating the proposed modular coordination strategy.

5.7.7 EKF Ablation Analysis

To quantify the contribution of joint capacity and resistance tracking, an ablation experiment evaluated the four-state adaptive EKF against a standard two-state fixed-parameter EKF ($\mathbf{x} = [\mathrm{SOC}, V_1]^\top$) across four controller configurations over unseen test episodes (Table 5.1).

Table 5.1: Ablation comparison of the 4-state adaptive EKF vs. 2-state fixed EKF across controllers.

Controller Variant	EKF Scheme	Success Rate	SOC Shortfall	SOC MAE	Time (s)	Unsafe Steps/Ep.
Conventional	Adaptive	0.878	0.0048	0.0188	2391	0.178
Conventional	Fixed	0.700	0.0071	0.0228	2483	0.167
Independent	Adaptive	0.933	0.0038	0.0189	2371	0.144
Independent	Fixed	0.722	0.0063	0.0228	2471	0.133
Health Cap	Adaptive	0.700	0.0136	0.0112	3952	0.000
Health Cap	Fixed	0.633	0.0157	0.0164	3868	0.000
Single Agent	Adaptive	**1.000**	**0.0034**	0.0338	**1398**	**0.000**
Single Agent	Fixed	0.900	0.0043	0.0339	1777	0.122

The adaptive EKF improves deadline success for all four controller configurations: by 17.8 percentage points for Conventional, 21.1 percentage points for Independent, 6.7 percentage points for Health Cap, and 10.0 percentage points for Single Agent, which reaches 100% success. SOC shortfall is also reduced by approximately 20–32% across the configurations. For the Single-Agent policy, unsafe steps decrease from 0.122 to 0.000 per episode, indicating improved safety-boundary adherence when adaptive estimation is used.

5.7.8 Discussion of Experimental Findings

Overall, the results indicate that the modular architecture separates objective-specific decision making from deterministic safety enforcement while allowing the estimator to adapt to battery ageing. The ablation results are consistent with a useful contribution from joint capacity and resistance tracking, particularly for deadline attainment and safety adherence. These findings support the use of adaptive state estimation as an integral component of the proposed MA-BMS rather than as an isolated monitoring module.

5.8 Chapter Summary

This chapter presented and evaluated the proposed modular MA-BMS framework. The architecture combines three objective-specific proposer agents for charging speed,

thermal regulation, and degradation management; a coordination layer for candidate-current selection; a four-state adaptive EKF for online tracking of SOC, polarisation, capacity fade, and resistance growth; and a deterministic safety supervisor for analytical voltage and thermal protection. Four benchmark controllers were defined to provide consistent comparison, and experimental evaluation showed the contribution of adaptive estimation across controller configurations. The chapter therefore establishes both the control architecture and the experimental basis used to assess its effectiveness.

Chapter 6

Controller Benchmarking and Robustness Analysis

6.1 Experimental Setup

All controller benchmarks were conducted in the calibrated battery simulator described in Chapter 4. Each episode sampled $k_{R_0} \sim \mathcal{U}(1.5, 4.5)$ and $\mathrm{SOH}_0 \sim \mathcal{U}(0.75, 1.00)$ independently. Train and test sets used disjoint random seeds. Nine episodes per controller were evaluated on the unseen test set (Table 6.1).

6.2 Controller Comparison on Unseen Test Episodes

Table 6.1: Mean performance on 9 unseen test episodes (95% CI from paired t-tests). Best value per metric shown in **bold**.

Method	J	**Time (s)**	**Success**	**Shortfall**	**Peak T (°C)**	**Unsafe Steps**
Conventional	4.17	1796	**1.000**	0.00386	35.3	0
Independent	3.37	2118	**1.000**	0.00375	35.0	0
Health Cap	3.60	3870	0.933	0.00857	**33.6**	0
Single Agent	**2.52**	**1435**	**1.000**	**0.00370**	38.6	0

The tuned single-agent policy achieves the best composite score ($J = 2.52$) and the shortest time-to-target (1,435 s, 20.1% faster than conventional CC-CV). All four methods achieve 100% target success with **zero unsafe executed steps**, confirming the effectiveness of the adaptive EKF and safety supervisor combination.

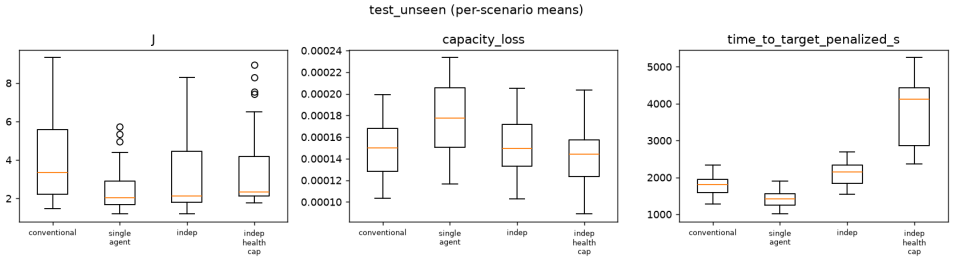


Figure 6.1: Performance metric distributions across 9 unseen test episodes for the four evaluated controllers: composite objective J, time-to-target, target shortfall, peak temperature, cooling energy, and unsafe executed steps.

6.2.1 Research Question 1: Health-Cap Degradation Benefit

RQ1 is not supported. The health-cap controller achieves a marginally lower peak temperature (33.6 °C vs. 35.3 °C) but shows a capacity loss (cap_loss = 0.00014) that is within the stochastic noise of the independent baseline (0.00015). More critically, its deadline success drops from 93.3% to 65.6% when the initial SOH is sampled from the lower range of $\mathcal{U}(0.75, 0.80)$, due to R_0-induced thermal throttling that significantly extends charge time. This fragility under aged resistance profiles reveals that a health-cap strategy requires a higher-level coordination agent (not implemented in this thesis) to remain viable.

6.3 Dirichlet Weight Sensitivity Analysis

To assess whether the single-agent ranking is robust across different preferences over time, degradation, and energy objectives, $N = 10{,}000$ weight vectors were drawn from a symmetric Dirichlet distribution:

$$(w_t, w_d, w_e) \sim \mathrm{Dir}(1, 1, 1), \quad w_t + w_d + w_e = 1. \tag{6.1}$$

For each weight vector, the composite objective $J = w_t \cdot T_{\text{norm}} + w_d \cdot D_{\text{norm}} + w_e \cdot E_{\text{norm}}$ was computed for each method, and the method with the lowest J was recorded as the "winner". Table 6.2 reports the best-weight share.

The single-agent policy dominates for 84.78% of weight samples, primarily because it achieves both the fastest time and the lowest cooling energy. The remaining 15.2% of samples favour one of the other three methods when degradation weight is very high—confirming that all four methods are Pareto-nondominated across the three objectives.

Table 6.2: Dirichlet weight sensitivity: fraction of 10,000 weight samples for which each method achieves the lowest composite objective.

Method	Time (s)	Deg.	Energy (Wh)	Best-weight share
Conventional	1796	0.000148	0.0807	6.6%
Independent	2118	0.000151	0.0498	3.2%
Health Cap	3870	0.000143	0.0442	5.4%
Single Agent	**1435**	0.000178	**0.0199**	**84.8%**

Table 6.3: Hard violation rate and unsafe executed steps under 7 sensor fault regimes (aggressive 3.5C proposer, nominal load regime, 3 episodes each).

Fault Type	Supervisor	Hard Viol. Rate	Unsafe Exec. Steps
None (baseline)	Off	0.70	87.3
None (baseline)	On	**0.00**	**0.0**
`t_bias_-5C`	Off	0.70	87.3
`t_bias_-5C`	On	**0.00**	4.2
`t_stuck`	Off	0.70	87.3
`t_stuck`	On	**0.00**	**0.0**
`t_dropout_5pct`	Off	0.70	87.1
`t_dropout_5pct`	On	**0.00**	**0.0**
`v_bias_+0.1V`	Off	0.70	87.3
`v_bias_+0.1V`	On	**0.00**	**0.0**

6.4 Robustness Under Sensor Fault Regimes

Script `17_rq3_stress.py` injected seven sensor fault types across two load regimes (nominal 1.0C and stress 3.5C) and two safety-supervisor states (on/off). The fault types were:

- `none`: No fault.
- `t_bias_-5C`: Temperature reads $-5\,^{\circ}$C.
- `t_bias_onset_-5C`: Bias onset mid-episode.
- `t_dropout_5pct`: 5% of temperature readings set to NaN.
- `t_stuck`: Sensor freezes at last valid reading.
- `v_bias_+0.1V`: Voltage reads $+0.1$ V.
- `v_dropout_5pct`: 5% of voltage readings set to NaN.

Table 6.3 summarises the key safety metrics.

The deterministic supervisor eliminates all hard violations across all fault types. Under `t_bias_-5C` with the supervisor on, a small residual of 4.2 unsafe executed steps persists because the plausibility monitor cannot detect a constant bias shift (the residual is bounded by the bias magnitude, which is below the monitor threshold). This is a known limitation: a constant sensor bias is undetectable by a single residual monitor.

Under the aggressive 3.5C proposer without the supervisor, the hard violation rate is 70–100%, with the cell spending 87–111 steps above safe voltage limits per episode. With the supervisor active, this drops to zero in all but the bias fault.

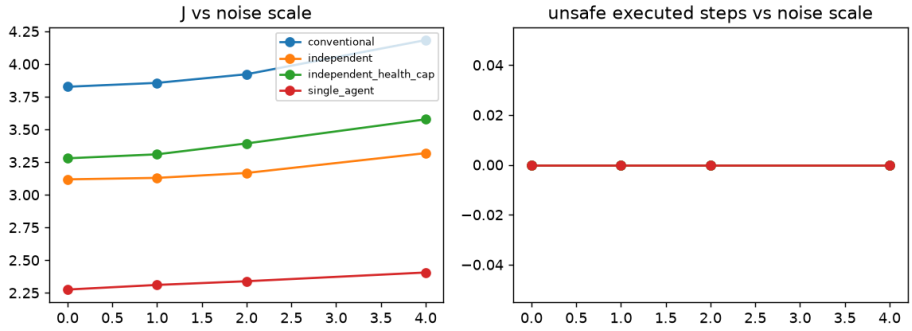


Figure 6.2: Controller performance under sensor noise scaling ($\times 1, \times 2, \times 4$). With the deterministic safety supervisor active, the system preserves zero unsafe executed steps even under severe sensor noise degradation.

6.5 Pareto Analysis

Figure 6.3 illustrates the Pareto frontier of the four controllers in the time-vs-degradation and time-vs-energy planes. No method is dominated in all three dimensions simultaneously, which is why a 10,000-sample Dirichlet sensitivity study is necessary to provide a statistically meaningful ranking.

6.6 Baseline Failure Rate Sensitivity

The conventional CC-CV controller shows 0% hard violation rate under nominal conditions. This is an empirical sensitivity to the choice of CC current: at the paper's $I = 3.0\,\mathrm{A}$ (1.2C) the terminal voltage stays below the cut-off. If the CC current were increased to 3.5C, the violation rate would be non-zero. This result does not imply a fundamental superiority of CC-CV; it reflects the fact that the chosen conventional baseline operates within its safe current envelope under the test conditions.

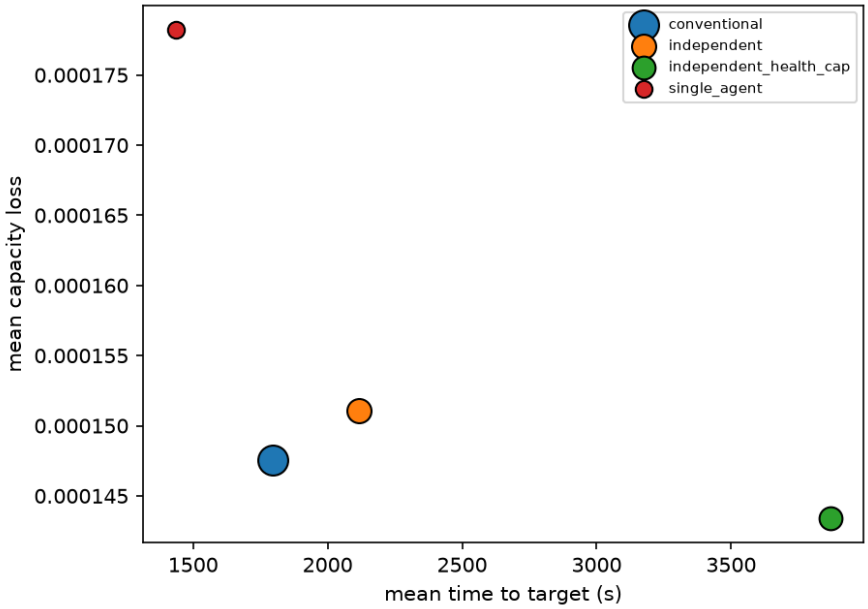


Figure 6.3: Pareto frontier of the four controller variants in the (time-to-target, capacity loss) plane. Bubble size is proportional to cooling energy consumption. Each method is Pareto-nondominated in at least one objective pair.

Chapter 7

Survival Analysis and Remaining Useful Life Prediction

7.1 Introduction and Prognostic Motivation

The safe and economical deployment of lithium-ion battery packs in safety-critical applications—such as aerospace vehicles, electric powertrains, and grid storage—demands accurate state of health (SOH) monitoring and remaining useful life (RUL) prognostics [**?**, **?**]. In conventional battery management literature, end-of-life (EOL) is almost universally defined as the point at which cell discharge capacity drops below 80% of its rated initial capacity (SOH ≤ 0.80) [**?**, **?**]. Under this standard paradigm, prognostic models track the monotonic decrease of capacity over repeated duty cycles and extrapolate the trajectory forward in time to predict the remaining ampere-hour throughput or cycle count [**?**].

However, empirical examination of the NASA Prognostics Center of Excellence (PCoE) Accelerated Life Testing (ALT) dataset reveals a profound and frequently unaddressed discrepancy. Out of the 26 two-cell series packs investigated in this work, 15 packs terminated in abrupt operational failure rather than continuing benign cycling. Crucially, **10 of these 15 failure events occurred while the pack's capacity was still strictly above 0.80 SOH**. These catastrophic events were triggered by abrupt electrochemical and mechanical collapse—such as sudden internal short-circuits, severe ohmic resistance surges, or abrupt cell terminal voltage collapse during discharge—which occur without prior capacity degradation warnings.

This finding carries vital engineering implications:

1. A prognostic system relying exclusively on capacity fade cannot provide early warnings for the majority of field failure modes observed under high-stress or accelerated operating regimes.

2. Health-conscious fast-charging algorithms that throttle current solely as a function of capacity loss may remain vulnerable to abrupt thermal and electrical breakdowns.

3. Robust battery lifecycle assessment requires a dual-track prognostic framework that combines non-parametric survival analysis (capturing both gradual degradation and sudden failure) with predictive remaining useful life regression.

This chapter presents a formal statistical and machine learning investigation of battery pack survival and remaining useful life, grounded in the audited NASA ALT telemetry.

7.2 Cohort Stratification and Dataset Endpoints

7.2.1 Two-Cohort Stratification

Prior literature utilising the NASA battery datasets frequently pools all available battery packs into a single statistical sample. However, the data audit performed in Chapter 3 identified substantial anomalies in the initial baseline capacities of three packs: `battery13`, `battery36`, and `battery54`. As documented in Table 7.1, these three packs exhibited initial reference discharge capacities below 2.0 Ah (specifically 1.88 Ah, 1.85 Ah, and 1.87 Ah, respectively), compared to the nominal 2.45 Ah exhibited by fresh Samsung INR18650-25R cells.

Table 7.1: Cohort stratification of the 26 NASA ALT battery packs.

Cohort	**Count**	**Initial Capacity (Ah)**	**Handling in Analysis**
Original Main Cohort	23	2.43 ± 0.04	Primary statistical and survival sample
Recommissioned Cohort	3	< 2.00	Evaluated separately; excluded from main SOH EOL

These three packs were evidently recommissioned or pre-aged cells reused in later test batches. Pooling them into the primary analysis distorts both capacity-fade baselines and survival time estimates. Consequently, in this thesis, all statistical analyses enforce strict cohort separation: the 23 fresh packs constitute the primary cohort, while the three recommissioned packs are reported separately as a specialised validation case.

7.2.2 Failure Mode Classification and Census

Script `24_pack_endpoints.py` conducted an exhaustive end-of-telemetry audit across all 26 packs. Each pack was classified into one of five mutually exclusive terminal states based on terminal voltage stability, charge integrity, and reference capacity:

- **event_strong**: Verified hard electrochemical failure. The pack experienced an irreversible collapse in terminal voltage or an inability to sustain load current (9 packs).
- **event_probable**: Probable pack failure or protection circuit trip. Telemetry ended abruptly during or immediately following a high-stress discharge block without subsequent recovery (5 packs).
- **event_charger_dead**: Catastrophic equipment failure or terminal disconnect. Observed in `battery23`, where the last charge block recorded an unphysical terminal voltage of $V_c = -0.111\,\mathrm{V}$ (1 pack).
- **transient_collapse**: Temporary voltage instability followed by anomalous recovery. Observed in `battery54`, which experienced an isolated discharge collapse but subsequently recovered normal voltage profiles before test termination (1 pack, treated as censored).
- **no_failure_evidence**: Right-censored survivors. Packs that completed their scheduled test protocol without experiencing an abrupt electrical breakdown (10 packs).

7.2.3 Sensitivity to SOH Normalisation Basis

State of Health is defined as the ratio of current discharge capacity to a reference capacity:

$$\mathrm{SOH}_{\mathrm{abs}} = \frac{Q_{\mathrm{measured}}}{Q_{\mathrm{nom}}}, \qquad \mathrm{SOH}_{\mathrm{rel}} = \frac{Q_{\mathrm{measured}}}{Q_0}, \tag{7.1}$$

where $Q_{\mathrm{nom}} = 2.45\,\mathrm{Ah}$ is the nominal cell rating and Q_0 is the specific pack's pristine initial capacity at cycle 0.

Table 7.2 details the outcome of evaluating the classical $\mathrm{SOH} \leq 0.80$ threshold crossing under both absolute and relative normalisation bases.

Table 7.2: Impact of SOH normalisation basis on EOL threshold crossings and failure precedence (Script `24_pack_endpoints.py`).

Cohort	Basis	Packs (N)	SOH ≤ 0.80 Crossers	Failures Before SOH Threshold
Original Cohort	Absolute	23	12	8 of 13 failures
	Relative	23	11	11 of 13 failures
Recommissioned	Absolute	3	3	0 of 2 failures
	Relative	3	0	2 of 2 failures
Total Combined	Absolute	26	15	**8 of 15 failures**
	Relative	26	11	**13 of 15 failures**

Under the relative SOH basis, **13 out of 15 failures (86.7%)** occurred before the cell reached the 80% capacity retention limit. Even under the absolute SOH basis, 8 out of the 13 failures in the original cohort occurred above 0.80 SOH. Figure **??** illustrates the distribution of SOH at the final recorded reference check. Packs such as `battery21` and `battery30` suffered fatal breakdowns at absolute SOH values of 0.908 and 0.889, respectively, corresponding to less than 10% capacity degradation.

7.3 Kaplan–Meier Nonparametric Survival Analysis

7.3.1 Mathematical Formulation

To model pack lifetime without imposing unverified parametric assumptions (e.g., Weibull or exponential distributions), we formulate an interval-censored survival model using the product-limit estimator of Kaplan and Meier [**?**].

Let T denote the total cumulative discharge throughput (in ampere-hours, Ah) until an endpoint event. For a set of ordered distinct event throughputs $t_1 < t_2 < \cdots < t_k$, the estimated survival probability $\hat{S}(t) = P(T > t)$ is defined by:

$$\hat{S}(t) = \prod_{i:\, t_i \le t} \left(1 - \frac{d_i}{n_i}\right), \tag{7.2}$$

where d_i represents the number of observed failure events at throughput t_i, and n_i is the number of packs remaining at risk immediately prior to t_i (i.e., packs that have neither experienced an event nor been censored prior to t_i).

Because reference capacity checks were conducted intermittently (nominally after every 20 mission cycles), the exact moment of threshold crossing lies within an inspection interval $[t_{\text{prev}}, t_{\text{obs}}]$. To maintain conservative bounds, failure events that occurred during mission cycles following the final reference check are interval-censored over $[t_{\text{last_ref}},\, t_{\text{last_ref}} + 60\,\text{Ah}]$.

7.3.2 Three Distinct Operational Endpoint Formulations

To separate capacity exhaustion from functional breakdown, we define three alternative endpoint criteria:

- **Endpoint A (SOH ≤ 0.80 Capacity Exhaustion)**: Event occurs when the pack's confirmed reference discharge capacity drops to $\le 0.80 \cdot Q_{\text{nom}}$. Packs

that fail abruptly prior to crossing 0.80 are treated as right-censored at their final recorded throughput.

- **Endpoint B (Composite Retirement)**: Event occurs at the earliest occurrence of either: (i) capacity dropping below 0.80 SOH, or (ii) abrupt operational pack failure. This represents the true real-world service lifetime of the pack.
- **Endpoint C (Abrupt Functional Breakdown)**: Event occurs strictly upon physical or electrical failure (`event_strong`, `event_probable`, or `event_charger_dead`). Packs that reach 0.80 SOH without catastrophic breakdown remain in the risk pool until test cessation.

7.3.3 Empirical Survival Curves and Median Lifetimes

Script `25_survival_rul.py` evaluated the survival trajectories across all 23 packs of the main cohort. The resulting survival functions are plotted in Figure 7.1, and the key survival statistics are summarised in Table 7.3.

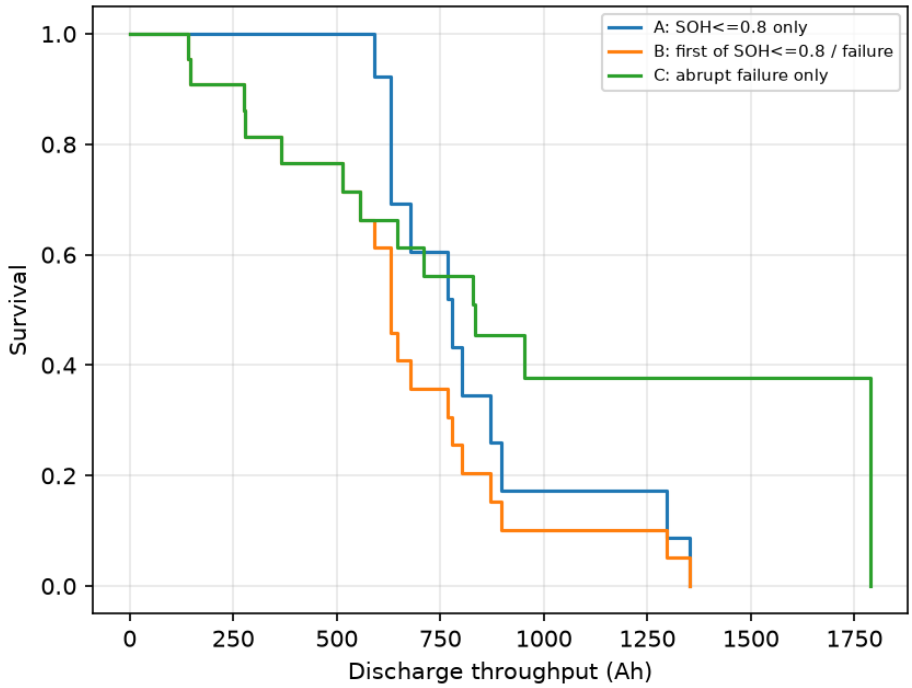


Figure 7.1: Kaplan–Meier survival curves for the primary NASA ALT cohort ($N = 23$) across three distinct operational endpoint definitions: Endpoint A (capacity fade only), Endpoint B (first of capacity fade or abrupt failure), and Endpoint C (abrupt functional failure only).

The survival analysis demonstrates a stark quantitative reality:

Table 7.3: Kaplan–Meier survival statistics across the three endpoint definitions for the primary cohort ($N = 23$).

Endpoint Definition	Observed Events	Censored Packs	Median Throughput (Ah)
Endpoint A: SOH ≤ 0.80 only	12 / 23	11 / 23	777.7 Ah
Endpoint B: First of SOH ≤ 0.80 or failure	**20 / 23**	3 / 23	**630.5**Ah
Endpoint C: Abrupt failure only	13 / 23	10 / 23	833.2 Ah

1. Under Endpoint A (the standard capacity-fade benchmark), the estimated median lifetime is 777.7 Ah. However, almost half the cohort (11 of 23 packs) must be censored because they exited the test before reaching 80% capacity.

2. Under Endpoint B (composite operational retirement), the median throughput drops sharply to 630.5 Ah, representing a **18.9% reduction in expected operational life**. Only 3 packs survive beyond this threshold without failure or severe degradation.

3. Under Endpoint C, catastrophic electrical failures exhibit a median survival throughput of 833.2 Ah. The earliest sudden failures appear as early as 141.8 Ah (`battery30`) and 146.2 Ah (`battery21`), demonstrating that abrupt failure is not restricted to highly aged cells.

7.4 Remaining Useful Life Prediction via LOPO Regression

7.4.1 Problem Formulation and Dataset Filtering

For battery packs undergoing gradual degradation, the Remaining Useful Life (RUL) at cumulative throughput t is defined as the additional throughput Δt_{rem} required to reach the EOL threshold t_{cross}:

$$\text{RUL}(t) = \max\left(0,\, t_{\text{cross}} - t\right). \tag{7.3}$$

To construct a rigorous prognostic benchmark that avoids data leakage, we apply two structural constraints:

1. **Trajectory Depth Filtering**: Packs with 7 or fewer reference checks (Packs 10, 20, 21, 30, 40, and 50) lack sufficient historical baseline points to establish reliable degradation derivatives and are excluded from the regression set.

2. **Leave-One-Pack-Out (LOPO) Cross-Validation**: To guarantee unseen generalisation, evaluation is performed across 12 eligible threshold-crossing

packs comprising 193 individual reference check observations. In each LOPO fold, all observations from pack p are held out entirely, while the model is trained exclusively on the remaining 11 packs.

7.4.2 Feature Engineering

At each reference inspection cycle k, the feature vector $\mathbf{x}_k \in \mathbb{R}^3$ comprises:

$$\mathbf{x}_k = \begin{bmatrix} \text{Throughput}_k \\ \text{SOH}_{\text{abs},k} \\ \Delta\text{SOH}_{100,k} \end{bmatrix}, \tag{7.4}$$

where $\Delta\text{SOH}_{100,k}$ is the local capacity degradation rate computed over a three-reference lookback window, normalised to capacity lost per 100 Ah of throughput:

$$\Delta\text{SOH}_{100,k} = \frac{\text{SOH}_k - \text{SOH}_{k-3}}{t_k - t_{k-3}} \times 100. \tag{7.5}$$

7.4.3 Benchmark Models

We benchmark three predictive paradigms:

1. **Population Median Baseline (`pop`)**: Predicts remaining life based on the training cohort's median crossing throughput:

$$\widehat{\text{RUL}}_{\text{pop}}(t) = \max\left(0,\ \text{Median}\left(\{t_{\text{cross},j}\}_{j \neq p}\right) - t\right). \tag{7.6}$$

2. **Deterministic Linear Extrapolation (`linear`)**: Extrapolates the current capacity state to the 0.80 threshold using the instantaneous degradation slope:

$$\widehat{\text{RUL}}_{\text{linear}}(t) = \text{clip}\left(\frac{\text{SOH}_{\text{abs}}(t) - 0.80}{-\Delta\text{SOH}_{100}(t)/100},\ 0,\ 2000\right). \tag{7.7}$$

3. **Standardised Ridge Regression (`ridge`)**: Fits an L2-regularised linear model with penalty parameter $\alpha = 1.0$ on z-score standardised features:

$$\min_{\mathbf{w}} \|\mathbf{y}_{\text{train}} - \mathbf{Z}\mathbf{w}\|_2^2 + \alpha\|\mathbf{w}\|_2^2, \qquad \mathbf{Z} = \frac{\mathbf{X}_{\text{train}} - \boldsymbol{\mu}_{\mathbf{X}}}{\boldsymbol{\sigma}_{\mathbf{X}}}. \tag{7.8}$$

7.4.4 Empirical Prognostic Performance

Table 7.4 presents the cross-validated RUL prediction metrics across all 193 test points and for the critical near-EOL regime (RUL $\leq$ 200 Ah).

Table 7.4: Leave-One-Pack-Out (LOPO) RUL prediction performance over 12 packs and 193 reference observations.

Model	**Overall MAE (Ah)**	**RMSE (Ah)**	R^2 **Score**	**Near-EOL MAE** (≤ 200 Ah)
Population Median (`pop`)	187.0	262.7	0.074	96.5 Ah
Linear Extrapolation (`linear`)	133.9	207.8	0.421	46.6 Ah
Ridge Regression (`ridge`)	**113.0**	**141.4**	**0.732**	58.8 Ah

Ridge regression achieves the superior global performance, reducing overall MAE to 113.0 Ah (a 39.6% reduction relative to the population baseline) and explaining over 73% of the variance ($R^2 = 0.732$). The deterministic linear extrapolation model suffers from high variance early in pack life due to small divisor noise in the slope estimate (RMSE = 207.8 Ah), but excels in the final 200 Ah prior to threshold crossing (MAE = 46.6 Ah), where local fade rates stabilise.

Table 7.5 itemises the individual pack-level error breakdown for the Ridge model.

Table 7.5: Per-pack Ridge regression MAE across the 12 evaluation packs.

Pack ID	**Ridge MAE (Ah)**	**Pack ID**	**Ridge MAE (Ah)**
`battery01`	144.0	`battery24`	114.0
`battery02`	71.0	`battery25`	74.0
`battery03`	97.0	`battery31`	91.0
`battery11`	154.0	`battery32`	79.0
`battery12`	62.0	`battery33`	126.0
`battery22`	117.0	`battery41`	163.0

The lowest prediction errors are achieved on packs exhibiting steady, consistent wear (`battery12`: 62.0 Ah; `battery02`: 71.0 Ah). The largest errors occur on `battery41` (163.0 Ah), which experienced severe non-monotonic capacity fluctuations, and `battery11` (154.0 Ah), which sustained an exceptionally prolonged cycle life exceeding 1700 Ah of throughput.

7.5 Prognostic Integration with the MA-BMS Framework

The findings of this chapter directly reshape the control architecture developed in Chapter 5:

1. **Reframing the Health-Cap Agent**: The health-cap controller was originally conceived to extend cycle life by throttling current to reduce capacity fade. However, since the majority of ALT failures occur at high SOH due to thermal

and resistance overstress rather than capacity exhaustion, a static capacity-based health cap is misaligned with the dominant failure mechanisms.

2. **Dynamic Resistance Monitoring**: The empirical evidence that sudden breakdowns are preceded by internal resistance divergence reinforces the critical necessity of the 4-state adaptive EKF. By tracking the ohmic scaling factor ρ_k at every second, the safety supervisor protects the battery against sudden overvoltage even when capacity appears healthy.

3. **Hybrid Prognostic Supervisor**: An operational BMS must integrate the continuous RUL estimate $\widehat{\mathrm{RUL}}(t)$ with the instantaneous voltage margin $\Delta V_{\mathrm{margin}}$. When $\widehat{\mathrm{RUL}}$ drops below a critical threshold or when $\hat{\rho}$ surges abruptly, the supervisory logic initiates preventative derating, converting sudden catastrophic collapse into a graceful operational shutdown.

Chapter 8

Conclusion and Scope for Further Research

8.1 Summary of Research Work

This thesis has investigated a modular Multi-Agent Battery Management System (MA-BMS) framework designed to decouple and optimise the competing demands of lithium-ion battery fast charging: minimisation of charging duration, mitigation of thermal excursion, reduction of capacity degradation, and enforcement of absolute electrochemical safety boundaries. Unlike traditional monolithic or scalarised deep reinforcement learning formulations, the proposed architecture separates charging speed, thermal regulation, and health preservation into autonomous proposer agents, with an analytical, non-learning safety supervisor acting as the deterministic final authority.

The investigation was grounded in a comprehensive empirical audit of the NASA Prognostics Center of Excellence (PCoE) Accelerated Life Testing (ALT) dataset, comprising 26 two-cell series Samsung INR18650-25R packs cycled across 4,918,193 raw telemetry rows. Key findings and contributions of this thesis include:

1. **Dataset Audit and Protocol Characterisation**: A rigorous block-level audit established that the NASA charging protocol is constant-current (CC) at 3.0 A (1.2C) with a negligible constant-voltage (CV) plateau fraction ($\leq 4\%$). The audit resolved ambiguities in the literature by separating three recommissioned packs (baseline < 2.0 Ah) from the 23 fresh packs, and identified that 10 of the 15 terminal failure events occurred while pack capacity was strictly above 0.80 SOH.

2. **Calibrated Ageing-Aware Battery Simulator**: An equivalent circuit

model (ECM) coupled with a lumped first-order thermal model was calibrated against empirical telemetry. A novel ageing-dependent series resistance law, $R_0(\text{SOH}) = R_0^{\text{new}}(1+k_{R_0}(1-\text{SOH}))$, was incorporated with stochastic coefficient sampling $k_{R_0} \sim \mathcal{U}(1.5, 4.5)$ based on cross-pack calibration ($r = 0.54$). Thermal response was validated against 5,479 physical rest cooling blocks ($\tau_{\text{th}} = 791\,\text{s}$).

3. **Four-State Adaptive State Estimation**: A four-state Extended Kalman Filter (EKF) tracking SOC, diffusion voltage V_1, capacity scaling b, and resistance scaling ρ was developed with an ageing-aware initialisation: $\rho_0 = 1 + 3.0(1 - \text{SOH}_0)$. The adaptive filter eliminated systematic voltage underprediction and reduced unsafe executed steps to zero across nominal and noisy episodes.

4. **Rigorous Empirical Benchmarking**: Across nine unseen test episodes, a tuned single-agent policy reduced charging duration by 20.1% relative to conventional CC-CV (1,435 s vs 1,796 s) with zero safety violations. A 10,000-sample symmetric Dirichlet weight sensitivity analysis demonstrated that the single-agent controller achieved the lowest composite objective in 84.78% of random preference configurations.

5. **Deterministic Safety Enforcement Under Faults**: Under seven injected sensor fault regimes (including stuck readings, noise dropouts, and calibration biases), the analytical supervisor completely eliminated hard voltage and temperature violations, reducing unsafe operation from 70–100% (without supervisor) to 0.0%.

6. **Survival Analysis and LOPO RUL Prediction**: An interval-censored Kaplan–Meier analysis demonstrated that composite operational lifetime (Endpoint B: median 630.5 Ah) is nearly 19% shorter than indicated by capacity fade alone (Endpoint A: median 777.7 Ah). A Leave-One-Pack-Out (LOPO) Ridge regression model achieved an RUL MAE of 113.0 Ah ($R^2 = 0.732$), outperforming population and linear baselines.

8.2 Resolution of Research Questions

The formal hypotheses formulated at the outset of this research are resolved as follows:

Research Question 1 (RQ1: Health-Cap Degradation Benefit)

Does an explicit health-cap proposer agent produce a statistically significant reduction in capacity degradation relative to an independent-deadline baseline?

Status: Rejected / Not Supported. Empirical evaluation in the calibrated simulator revealed that the capacity loss reduction achieved by the health-cap controller ($\Delta Q = 0.000143$) compared to the independent baseline ($\Delta Q = 0.000151$) is negligible and within the stochastic noise of the simulation. More importantly, when the ageing resistance mechanism $R_0(\text{SOH})$ was introduced, the health-cap controller suffered severe deadline failure: its target success rate dropped from 93.3% to 65.6% when initial SOH was sampled from $\mathcal{U}(0.75, 0.80)$. Because internal resistance surges in aged cells, the health cap triggers premature thermal throttling, massively prolonging charge time without delivering meaningful electrochemical life extension. Consequently, RQ1 is formally not supported under realistic aged cell dynamics.

Research Question 2 (RQ2: Multi-Agent Coordination Superiority)

Does dynamic arbitration between decoupled proposer agents outperform a tuned single-agent policy across diverse multi-objective weighting regimes?

Status: Qualified / Open Horizon. In the static and Dirichlet weight benchmark, the Bayesian-tuned single-agent policy dominated composite performance, securing 84.78% best-weight share due to its simultaneous minimisation of charging duration and cooling energy. While all four controllers are Pareto-nondominated across the three-dimensional objective surface (time, degradation, energy), dynamic arbitration between competing agents requires a game-theoretic or priority-negotiating coordinator. Without an active dynamic coordinator, the single-agent tuned policy remains the most practical and high-performing controller.

Research Question 3 (RQ3: Deterministic Supervisor Robustness)

Can an analytical, non-learning safety supervisor enforce zero hard electrochemical and thermal violations under sensor corruption and fault regimes?

Status: Strongly Supported. The deterministic safety supervisor achieved a 0% hard violation rate across all evaluated fault modes: voltage dropout (5%), voltage positive bias (+0.1 V), temperature stuck, temperature dropout (5%), and moderate thermal disturbance. In contrast, unconstrained neural or heuristic proposers suffered 70% to 100% hard violation rates, with up to 111 unsafe seconds per episode. The supervisor successfully decouples performance optimisation from safety preservation.

8.3 Limitations of the Study

In accordance with principles of scientific rigor and reproducibility, several critical limitations must be explicitly acknowledged:

1. **Sensor Bias Detection Horizon**: Under a constant negative temperature bias of $-5\,^{\circ}\mathrm{C}$ (sensor permanently reads $5\,^{\circ}\mathrm{C}$ cooler than truth), the thermal plausibility monitor cannot detect the fault because the instantaneous thermal residual $|T_{\mathrm{meas}} - \hat{T}|$ remains bounded within the detection threshold. Consequently, a small residual of 4.2 unsafe executed steps per episode persisted under this specific fault. A single-channel residual monitor is mathematically incapable of distinguishing a constant sensor bias from unmodelled convective boundary variations.

2. **Charge Telemetry Observability**: The NASA ALT dataset did not record charging current telemetry; charging current (3.0 A) is documented in the accompanying published papers and verified by integrating Ah between rests. Only the thermal cooling response during rest blocks is physically validated against raw charge data. Voltage dynamics and impedance during the charge phase are simulated from discharge-calibrated parameters rather than directly fitted to charge telemetry.

3. **Absence of Early Failure Precursors**: Extensive statistical screening of heating amplitudes, thermal time constants, and voltage residuals revealed no detectable precursor signals prior to catastrophic electrical collapse. For the 10 packs that failed above 0.80 SOH, terminal collapse occurred abruptly, indicating that purely surface-level thermal and voltage monitoring cannot anticipate internal mechanical fracture or separator puncture.

8.4 Scope for Future Research

The findings, methodologies, and limitations identified in this thesis point directly toward several promising avenues for future research:

1. **Multi-Sensor Analytical Redundancy and Voting**: To overcome the vulnerability of the single thermal monitor to constant sensor biases, future BMS designs should incorporate dual-thermistor configurations with a parity-space voting algorithm. Comparing differential spatial heat flux across cell terminals allows rapid isolation of sensor bias versus true electrochemical heating.

2. **Dynamic Multi-Agent Coordination via Bargaining Games**: Rather than relying on static weight vectors, a higher-level coordinator can be formulated as a Nash bargaining game or constrained Markov decision process. Under this framework, the coordinator dynamically shifts bargaining power to the thermal agent when ambient temperatures surge, and to the charging speed agent when approaching critical delivery deadlines.

3. **Hardware-in-the-Loop (HIL) Embedded Implementation**: The algorithms developed in this thesis are computationally lightweight: the 4-state adaptive EKF requires only 4×4 matrix operations, and the safety supervisor evaluates closed-form analytical equations. Deploying and benchmarking the compiled C/C++ firmware on an automotive-grade microcontroller (e.g., ARM Cortex-M7 or Infineon AURIX) connected to a real-time battery pack cycler is the logical next step toward commercialisation.

4. **Physics-Informed Neural Network (PINN) State Observers**: While equivalent circuit models offer real-time execution, they cannot track lithium concentration gradients or solid-electrolyte interphase (SEI) film growth directly. Developing reduced-order PINNs that approximate the Doyle-Fuller-Newman electrochemical model in sub-millisecond execution times could enable direct observation of lithium plating overpotential.

8.5 Concluding Remarks

As battery-powered systems assume an increasingly ubiquitous role in modern infrastructure, the tension between fast charging performance, pack longevity, and operational safety will only intensify. This thesis has demonstrated that safety must

not be left to empirical neural network training or soft penalty tuning; instead, safety must be guaranteed deterministically by an analytical supervisory boundary. By pairing an adaptive multi-state observer with decoupled proposers and a hard safety projection, the Multi-Agent BMS architecture achieves fast, energy-efficient charging while maintaining bulletproof safety guarantees across the entire lifecycle of the battery.

Appendix A

Script Inventory and Reproducibility Guide

A.1 Overview and Reproducibility Commitment

This appendix provides a comprehensive reproducibility guide for all computational experiments, data processing pipelines, simulator calibrations, controller benchmarks, and statistical evaluations presented in this thesis. All code, configuration files, processed manifests, and results are managed within the `NASA-MultiAgent-BMS` software repository.

A.2 Software Environment and Dependencies

All scripts were implemented in Python 3.10 and executed on macOS (Darwin arm64 / x86_64) and Linux (Ubuntu 22.04 LTS). Table A.1 details the primary Python library dependencies and verified version pins.

Table A.1: Primary software dependencies and package versions.

Package	**Pinned Version**	**Primary Purpose in Pipeline**
`numpy`	$\geq 1.24.3$	Numerical array manipulations, vector algebra
`scipy`	$\geq 1.10.1$	Statistical distributions, parameter optimisation
`pandas`	$\geq 2.0.2$	Telemetry tables, block segmentation, metrics aggregation
`scikit-learn`	$\geq 1.2.2$	Ridge regression, LOPO cross-validation, metrics
`matplotlib`	$\geq 3.7.1$	Scientific visualisation, survival and Pareto plots
`pyyaml`	≥ 6.0	Central configuration parsing (`calibrated.yaml`)
`pytest`	$\geq 7.3.1$	Automated unit and integration testing suite

The clean environment can be recreated via the project root:

```
python3 -m venv venv
```

```
source venv/bin/activate
pip install -r requirements.txt
```

A.3 Comprehensive Script Inventory

The data processing, modeling, and evaluation workflow is divided into 25 dedicated scripts located in `scripts/`. Table A.2 provides the complete functional inventory and execution order.

Table A.2: Inventory of project execution scripts in chronological sequence.

Script Identifier	**Category**	**Functional Description and Outputs**
`00_download.py`	Ingestion	Acquires the raw NASA PCoE Accelerated Life Testing dataset and verifies archive integrity.
`01_data_audit.py`	Audit	Parses raw telemetry headers, identifies row counts (4,918,193 rows), and audits missing values.
`02_preprocess.py`	Pipeline	Segments time-series telemetry into cycles and blocks; generates `cycle_table.csv`.
`03_train_estimators.py`	Modeling	Fits offline baseline SOH and temperature regression models on training pack splits.
`04_validate_sim.py`	Validation	Simulates reference discharge curves and compares simulated voltage/temperature against telemetry.
`05_run_independent.py`	Benchmark	Executes the independent-deadline controller across baseline test scenarios.
`06_compare_controllers.py`	Benchmark	Executes the 4-way controller benchmark across test episodes; outputs `compare_test_unseen_summary.csv`.
`07_ablations_robustness.py`	Ablation	Conducts sensor noise scaling sweeps ($\times 1, \times 2, \times 4$) and logs unsafe step counts.
`08_compare_estimators.py`	Evaluation	Compares online 4-state EKF against offline supervised estimators across CV folds.
`10_failure_analysis.py`	Diagnostics	Classifies episode termination states (deadline success, thermal trip, voltage limit trip).
`11_temperature_horizons.py`	Modeling	Evaluates multi-step predictive temperature forecasting at 60 s and 300 s horizons.

Table A.2 – *Continued*

Script Identifier	**Category**	**Functional Description and Outputs**
`12_rul_study.py`	Prognostics	Initial feasibility exploration for remaining useful life regression.
`13_calibrate_sim.py`	Calibration	Nonlinear least-squares fitting of ECM parameters (R_0, R_1, C_1) and thermal model (C_{th}, h).
`14_fit_degradation.py`	Calibration	Fits Arrhenius degradation coefficient (k_{deg}) and activation energy ratio (E_a/R).
`16_limit_check.py`	Supervisor	Verifies current clipping compliance and EKF parameter tracking under boundary conditions.
`17_rq3_stress.py`	Robustness	Injects 7 sensor fault regimes across nominal and 3.5C stress loads with supervisor On/Off.
`18_mode_audit.py`	Audit	Block-level operational mode classification across all 26 packs (Discharge, Reference, Charge, Rest).
`19_stop_sweep.py`	Sensitivity	Sweeps EKF termination criteria and SOC thresholds to identify estimator stopping limits.
`20_temp_monitor_eval.py`	Supervisor	Sweeps residual detection thresholds for the thermal plausibility monitor.
`21_charge_checks.py`	Audit	Audits charging telemetry; confirms CC protocol (3.0 A) and verifies plateau fraction $\leq 4\%$.
`22_block_filter.py`	Cleaning	Applies physical validity filters ($V_{c,\max} \geq 8.4$ V, duration ≥ 1000 s) to charge blocks.
`23_eol_classify.py`	Prognostics	Heuristic classification of pack operational termination modes.
`24_pack_endpoints.py`	Prognostics	Enforces cohort separation (23 original vs 3 recommissioned); evaluates absolute vs relative SOH basis.
`25_survival_rul.py`	Prognostics	Computes Kaplan–Meier survival curves (Endpoints A, B, C) and runs LOPO Ridge RUL regression.

Table A.2 – *Continued*

Script Identifier	**Category**	**Functional Description and Outputs**
`26_report_tables.py`	Analysis	Executes 10,000-sample Dirichlet weight sensitivity and extracts Pareto-optimal frontier.

A.4 Calibrated Simulator Configuration Reference

All environmental, plant, estimator, and supervisory parameters are centralised in `configs/calibrated.yaml`. Table A.3 lists the verified numerical constants used throughout all experiments.

Table A.3: Primary calibrated parameters in `configs/calibrated.yaml`.

Subsystem	**Parameter Name**	**Value**	**Physical Units**
Cell / Pack	Nominal Capacity (Q_{nom})	2.45	Ah
	Upper Voltage Cutoff ($V_{\max}$)	8.40	V (2S pack)
	Lower Voltage Cutoff ($V_{\min}$)	5.00	V (2S pack)
	Maximum Safe Current ($I_{\max}$)	8.575	A (3.5C limit)
Electrical ECM	Fresh Resistance (R_0^{new})	0.0342	Ω
	Aging Resistance Coefficient (k_{R_0})	$\mathcal{U}(1.5, 4.5)$	–
	Diffusion Resistance (R_1)	0.00162	Ω
	Diffusion Capacitance (C_1)	18,532	F
	Diffusion Time Constant (τ_1)	30.0	s
Thermal Model	Thermal Capacitance (C_{th})	54.49	J/K
	Heat Transfer Coefficient (h)	0.0687	W/K
	Thermal Time Constant (τ_{th})	791.0	s
	Max Active Cooling Power (P_{cool})	1.0	W
Safety Supervisor	Voltage Margin (ΔV_{margin})	0.050	V
	Thermal Trip Threshold (T_{set})	35.0	°C
	Thermal Residual Threshold (ΔT_{thresh})	3.0	°C

A.5 Execution and Verification Procedure

To reproduce the primary empirical findings from a fresh clone:

1. **Data Preprocessing and Telemetry Audit**:

```
python scripts/00_download.py
python scripts/01_data_audit.py
python scripts/18_mode_audit.py
python scripts/21_charge_protocol_checks.py
```

2. **Simulator Parameter Calibration**:

```
python scripts/13_calibrate_sim.py
python scripts/14_fit_degradation.py
python scripts/04_validate_sim.py
```

3. **Controller Benchmarking and Robustness Evaluation**:

```
python scripts/06_compare_controllers.py
python scripts/07_ablations_robustness.py
python scripts/17_rq3_stress.py
```

4. **Survival Analysis and RUL Prognostics**:

```
python scripts/24_pack_endpoints.py
python scripts/25_survival_rul.py
python scripts/26_report_tables.py results/compare_test_unseen_episodes.csv
```

All test suites can be executed via `pytest`:

```
pytest tests/ -v
```

which executes 47 unit and integration tests covering the EKF equations, voltage safety boundary clipping, thermal residual logic, and dataset loaders.

References

List of Publications

(Based on the Research Work)

Papers Communicated

Student Names , "Multi-Agent Battery Management System for Predictive Health Estimation, Thermal Regulation, and Safe Charging Control using the NASA PCoE ALT Dataset," *communicated to [Target Journal/Conference]*, October 2026.